Time Series Analysis: 780 (Tampere University)

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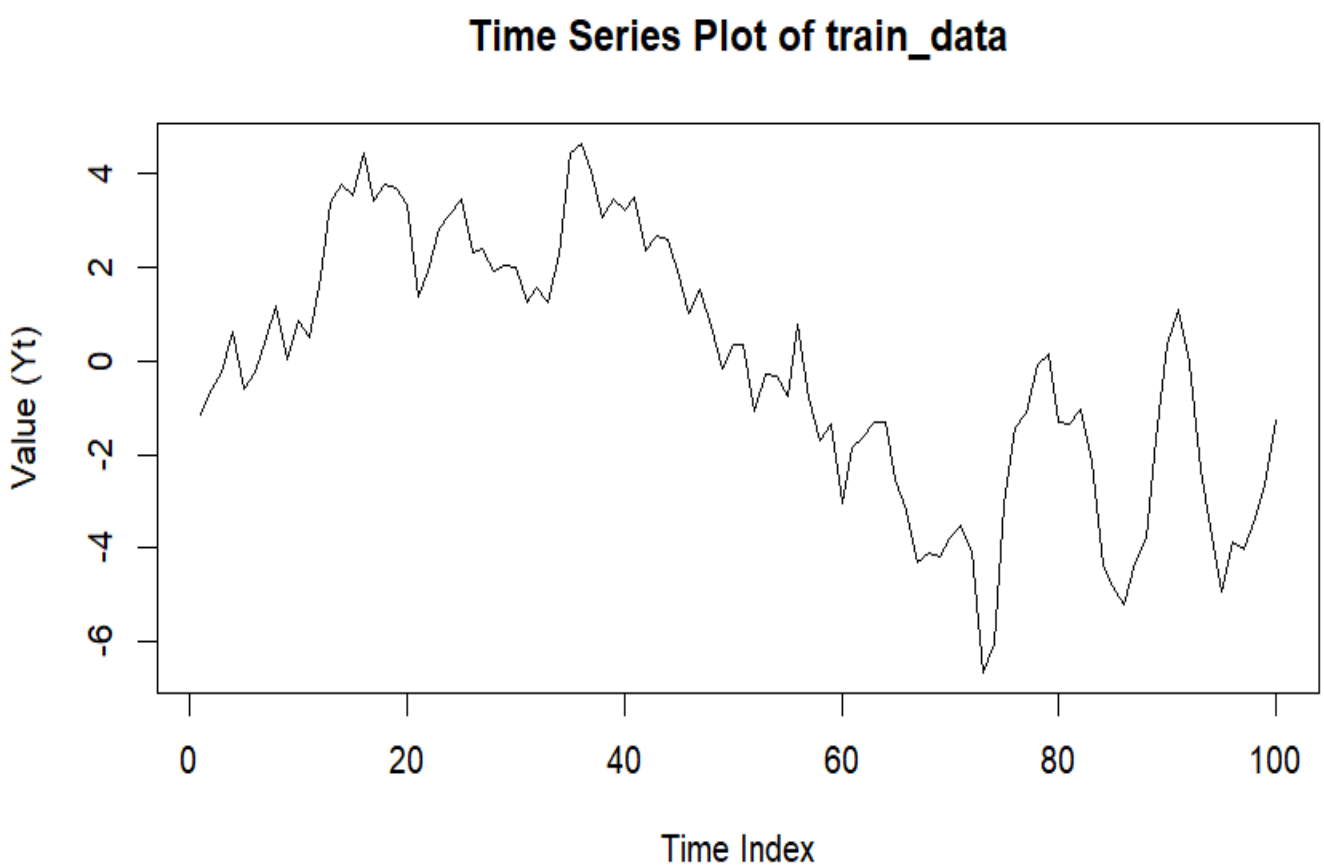
Column Assigned: V45

Assignment 1

Step 1: Preliminary analysis of Orders

1.Plot of the time series Y_t

```
# Load required package
  install.packages("tseries")
  library(tseries)
# Read dataset
  Full_Dataset <- read.csv("Case_study.csv")
# Select first 100 observations for training
  train_data <- Full_Dataset[1:100, 45]
# Convert to time series object
  yt_series <- ts(train_data)
# Plot the time series
  plot(yt_series,
       main = "Time Series Plot of train_data",
       ylab = "Value (Yt)",
       xlab = "Time Index")
```

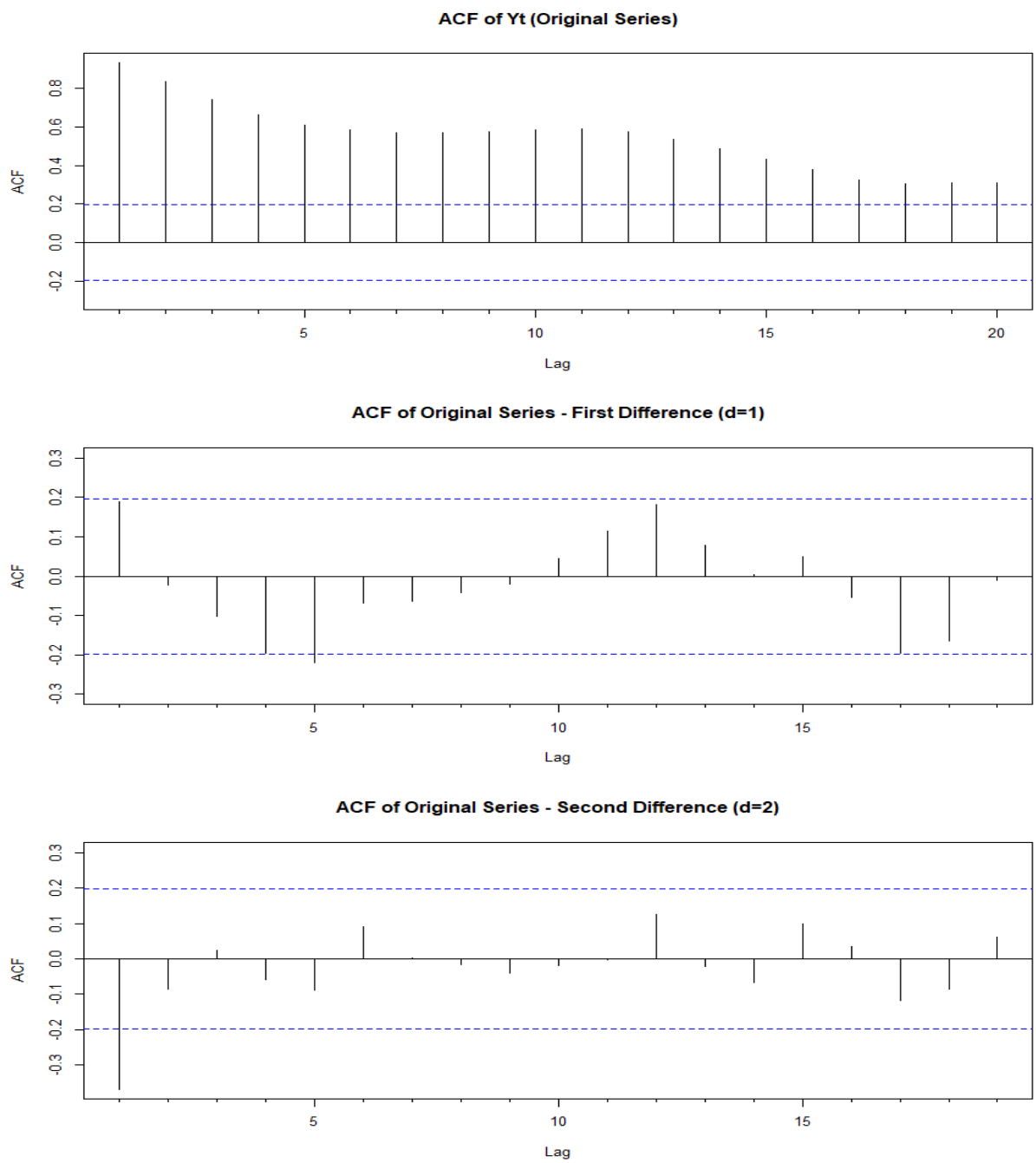


From the time series plot of the first 100 observations, the series does not fluctuate around a constant mean. It first increases and then shows a clear decreasing movement. The level of the series changes over time, so visually it does not look stationary.

2. Analysis of d.

(a): Display the autocorrelation function of (Yt) and ∇kYt, k = 1, 2 where ∇Yt = Yt − Yt−1.

```
# Load forecasting package
  install.packages("forecast")
  library(forecast)
# Compute first and second differences
  diff1_yt <- diff(yt_series, differences = 1)
  diff2_yt <- diff(yt_series, differences = 2)
# Plot ACF for original series and differenced series
  par(mfrow=c(3,1))
  Acf(yt_series, main="ACF of Yt (Original Series)")
  Acf(diff1_yt, main="ACF of Original Series - First Difference (d=1)")
  Acf(diff2_yt, main="ACF of Original Series - Second Difference (d=2)")
```



(1) What do you observe?

- For the original series Y_t, the ACF values are positive and relatively large for many lags, and they decay slowly.
- For the first difference (d = 1), most of the ACF bars are inside the confidence bounds, except one clear negative spike at lag 5.
- For the second difference (d = 2), almost all autocorrelations are inside the confidence interval.

(2) Using this informal graphical procedure, is there some evidence that the series (Yt) might be non-stationary?

Yes, there is some evidence of non-stationarity. The ACF of the original series decreases slowly and remains significant for many lags. This kind of behaviour is typical for a non-stationary process with a possible unit root.

(b) Implement a unit root test, the so-called Augmented DickeyFuller test (ADF).

```
# Augmented Dickey-Fuller test for stationarity
  adf_result <- adf.test(yt_series, alternative = "stationary")
  print(adf_result)

        Augmented Dickey-Fuller Test

 data:  yt_series
 Dickey-Fuller = -3.291, Lag order = 4, p-value = 0.07663
 alternative hypothesis: stationary
```

From the R output:

- Dickey-Fuller statistic = -3.291
- p-value = 0.07663

Since the p-value is greater than 0.05, we do not reject the null hypothesis at the 5% significance level. Therefore, there is not enough strong evidence to conclude that the series is stationary.

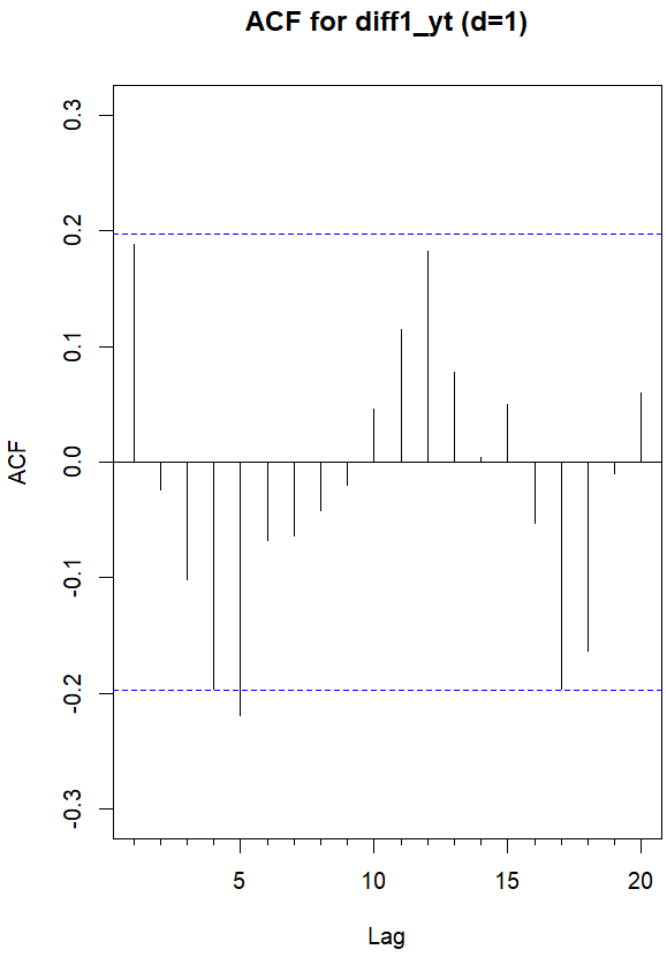
(c) Which order do you suggest for d?

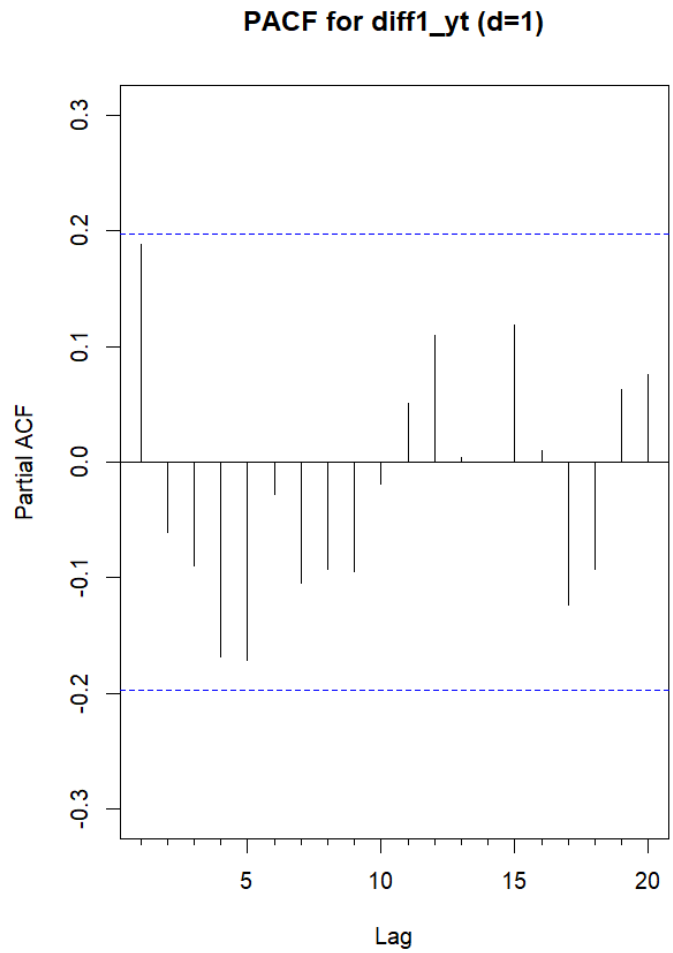
Based on the graphical analysis and the ADF test result, one differencing seems sufficient to remove the non-stationarity. Therefore, I suggest:

$$d = 1$$

3. Analysis of (p, q):

```
# Analyze AR and MA orders using ACF and PACF
  par(mfrow=c(1,2))
  Acf(diff1_yt, main="ACF for diff1_yt (d=1)", lag.max=20)
  Pacf(diff1_yt, main="PACF for diff1_yt (d=1)", lag.max=20)
```





Since $d = 1$, we analyze the ACF and PACF of the first differenced series.

- From the ACF of ∇Y_t, there is one clear significant spike at lag 5. Therefore, a reasonable upper bound for the moving average part is:

$$q_{max} = 5$$

- From the PACF of ∇Y_t, all spikes are inside the confidence bounds and there is no clear cut off pattern. So a reasonable upper bound for the autoregressive part is:

$$p_{max} = 1$$

Step 2: Estimation and Selection of ARIMA Models

1. Compute the AIC and BIC for the 16 remaining models.

```
# Create container to store model results
  model_results <- data.frame(Model = character(), AIC = double(), BIC = double(), string
sAsFactors = FALSE)
# Fit ARIMA(p,1,q) models for p,q = 1,...,4
  for (p in 1:4) {
    for (q in 1:4) {

      # Fit model
      fit <- Arima(yt_series, order = c(p, 1, q))

      # Store AIC and BIC
      model_name <- paste0("ARIMA(", p, ",1,", q, ")")
      model_results <- rbind(model_results, data.frame(
        Model = model_name,
        AIC = fit$aic,
        BIC = fit$bic
      ))
    }
  }
# Display comparison table
  print(model_results)
          Model      AIC      BIC
1  ARIMA(1,1,1) 285.3009 293.0862
2  ARIMA(1,1,2) 283.4853 293.8658
3  ARIMA(1,1,3) 284.4887 297.4643
4  ARIMA(1,1,4) 285.1367 300.7074
5  ARIMA(2,1,1) 281.7994 292.1799
6  ARIMA(2,1,2) 280.4872 293.4628
7  ARIMA(2,1,3) 281.6558 297.2265
8  ARIMA(2,1,4) 283.5566 301.7224
9  ARIMA(3,1,1) 288.9452 301.9208
10 ARIMA(3,1,2) 285.3911 300.9619
11 ARIMA(3,1,3) 283.3737 301.5395
12 ARIMA(3,1,4) 285.6390 306.4000
13 ARIMA(4,1,1) 283.3567 298.9274
14 ARIMA(4,1,2) 285.8132 303.9791
15 ARIMA(4,1,3) 284.4929 305.2538
16 ARIMA(4,1,4) 284.3442 307.7003
```

At this step, we assume that the differencing order is already determined as $d = 1$.
Also, we consider $1 \leq p \leq 4$and $1 \leq q \leq 4$, excluding the cases (0,0), (p,0), and (0,q). Therefore, 16 ARIMA models were estimated and compared using AIC and BIC

After fitting all 16 models, the information criteria were computed.

- The **minimum AIC** value was obtained for **ARIMA(2,1,2)**: AIC = 280.4872
- The **minimum BIC** value was obtained for **ARIMA(2,1,1)**: BIC = 292.1799

Since AIC and BIC have different statistical properties (AIC focuses more on efficiency while BIC is more consistent), it is possible that they select different models, which actually happened here.

In addition, we also consider a model that is close to the minimum AIC value, namely **ARIMA(2,1,3)**, because its AIC is very close to the best one.

2. Estimates of the best three specifications

```
# Fit selected best models
  model_A <- Arima(yt_series, order = c(2, 1, 2)) # Best AIC
  model_B <- Arima(yt_series, order = c(2, 1, 1)) # Best BIC
  model_C <- Arima(yt_series, order = c(2, 1, 3)) # Adjacent model
# Print parameter estimates
  cat("--- Estimates for ARIMA(2,1,2) [Best AIC] ---\n")
  --- Estimates for ARIMA(2,1,2) [Best AIC] ---
  print(model_A$coef)
       ar1        ar2        ma1        ma2
  1.6746930 -0.9280944 -1.6311279  0.8208433
  cat("AIC:", model_A$aic, " | BIC:", model_A$bic, "\n\n")
  AIC: 280.4872  | BIC: 293.4628

  cat("--- Estimates for ARIMA(2,1,1) [Best BIC] ---\n")
  --- Estimates for ARIMA(2,1,1) [Best BIC] ---
  print(model_B$coef)
       ar1        ar2        ma1
  1.0100452 -0.2878323 -0.8455775
  cat("AIC:", model_B$aic, " | BIC:", model_B$bic, "\n\n")
  AIC: 281.7994  | BIC: 292.1799

  cat("--- Estimates for ARIMA(2,1,3) [Adjacent] ---\n")
  --- Estimates for ARIMA(2,1,3) [Adjacent] ---
  print(model_C$coef)
       ar1        ar2        ma1        ma2        ma3
  1.7099064 -0.9466604 -1.6010616  0.6952268  0.1066335
  cat("AIC:", model_C$aic, " | BIC:", model_C$bic, "\n\n")
  AIC: 281.6558  | BIC: 297.2265
```

The estimated coefficients for the three selected models are presented below.

Model	ar1	ar2	ma1	ma2	ma3	AIC	BIC
ARIMA(2,1,2)	1.674693	-0.928094	-1.631128	0.820843	—	280.4872	293.4628
ARIMA(2,1,1)	1.010045	-0.287832	-0.845578	—	—	281.7994	292.1799
ARIMA(2,1,3)	1.709906	-0.946660	-1.601062	0.695227	0.106634	281.6558	297.2265

These three models were selected as follows:
ARIMA(2,1,2) minimizes AIC.
ARIMA(2,1,1) minimizes BIC.
ARIMA(2,1,3) has an AIC value very close to the minimum and is therefore considered an adjacent model.

Step 3: Diagnostic tests

1. LjungBox test for the first 10 lags. Also, provide ACF and PACF plots.

```
# Store models in a list
  models <- list(
    "ARIMA(2,1,2)" = model_A,
    "ARIMA(2,1,1)" = model_B,
    "ARIMA(2,1,3)" = model_C
  )
# Residual diagnostics
  for (name in names(models)) {

    cat("\n==============================================\n")
    cat("Diagnostic Tests for", name, "\n")

    res <- residuals(models[[name]])

    # Ljung-Box test for autocorrelation
    lb_test <- Box.test(res, lag = 10, type = "Ljung-Box")
    print(lb_test)

    # Plot ACF and PACF of residuals
    par(mfrow = c(1,2))
    Acf(res, main = paste("ACF of Residuals -", name))
    Pacf(res, main = paste("PACF of Residuals -", name))
  }
```

```
==============================================
Diagnostic Tests for ARIMA(2,1,2)

        Box-Ljung test

data:  res
X-squared = 4.5412, df = 10, p-value = 0.9196


==============================================
Diagnostic Tests for ARIMA(2,1,1)

        Box-Ljung test

data:  res
X-squared = 2.6987, df = 10, p-value = 0.9877


==============================================
Diagnostic Tests for ARIMA(2,1,3)

        Box-Ljung test

data:  res
X-squared = 3.3861, df = 10, p-value = 0.9708
```

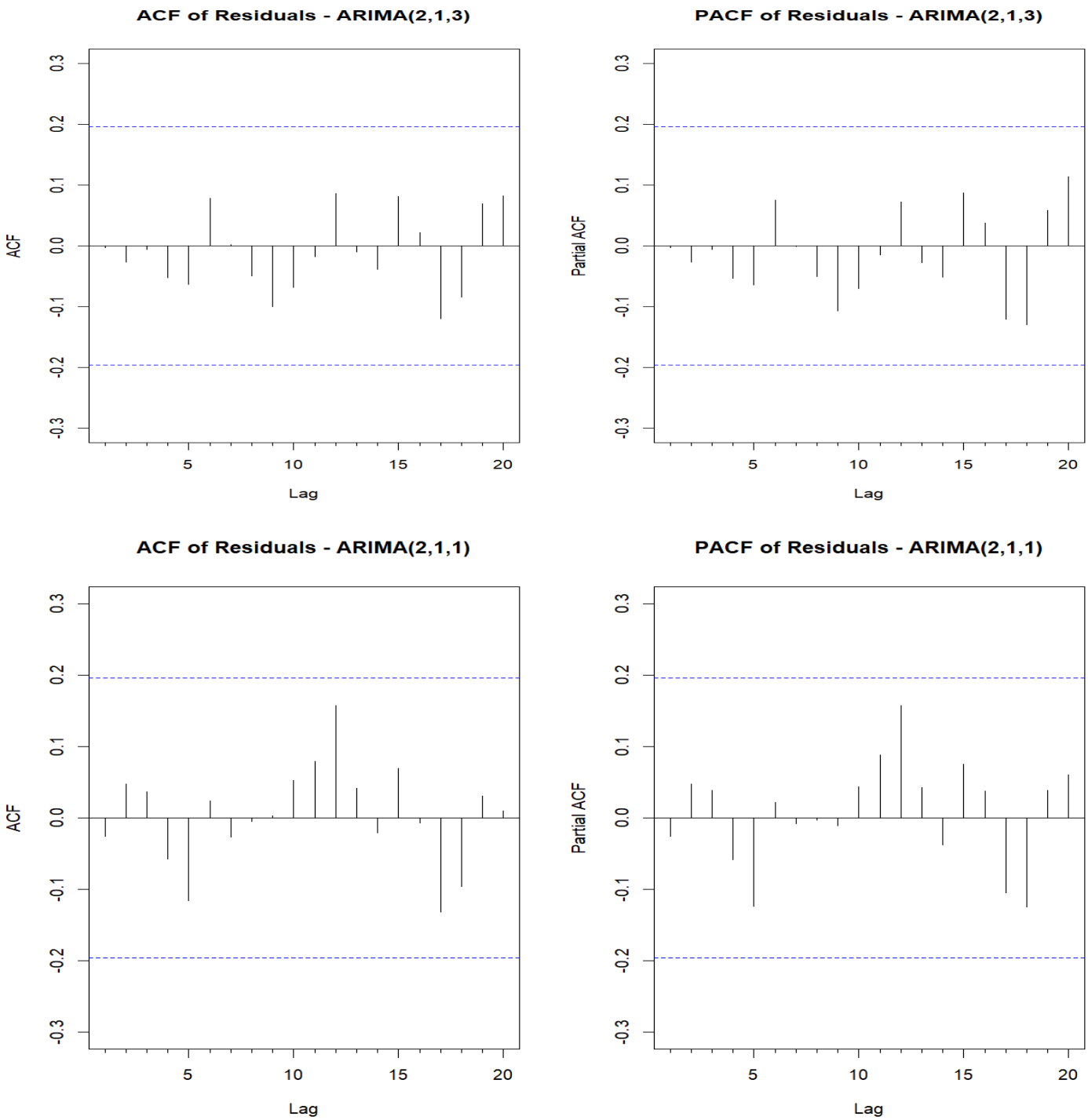
At this stage, I evaluate whether the selected ARIMA models adequately capture the dependence structure of the data by examining the residuals.

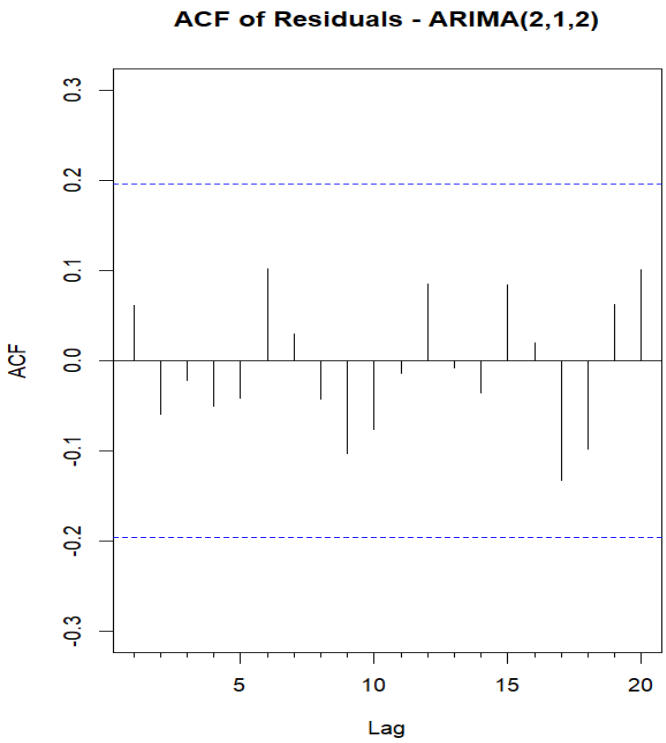
The Ljung–Box test results are:

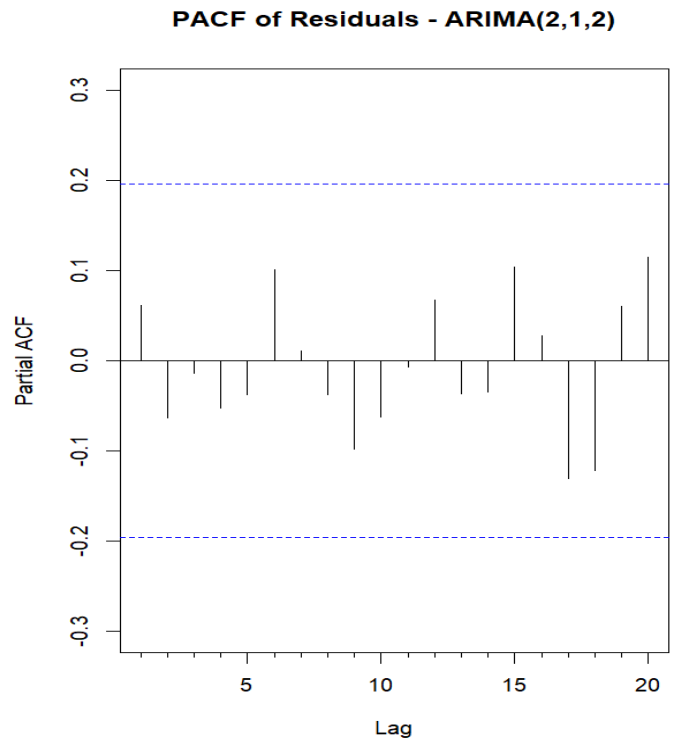
ARIMA(2,1,2): p-value = 0.9196
ARIMA(2,1,1): p-value = 0.9877
ARIMA(2,1,3): p-value = 0.9708

All p-values are much larger than 0.05. Therefore, I do not reject the null hypothesis of no serial correlation. This indicates that the residuals do not show significant autocorrelation up to lag 10.

ACF and PACF of Residuals:







From the ACF and PACF plots of the residuals, I observe that:

- ✓ Almost all spikes lie within the 95% confidence bounds.
- ✓ There are no clear significant spikes.
- ✓ No systematic pattern appears across lags.
- ✓ The residuals fluctuate randomly around zero.
- ✓ The behavior of the residual autocorrelations looks very similar to white noise.

Conclusion:

Based on both the Ljung–Box test results and the ACF/PACF plots of the residuals, there is no evidence of remaining autocorrelation. This suggests that all three ARIMA models are well specified and adequately capture the dynamics of the time series.
Among them:
ARIMA(2,1,2) is preferred according to AIC.
ARIMA(2,1,1) is preferred according to BIC.
Overall, the residual diagnostics support the validity of the selected models.

2. Normality of Residuals

```
# Normality check for residuals
  models <- list(
    "ARIMA(2,1,2)" = model_A,
    "ARIMA(2,1,1)" = model_B,
    "ARIMA(2,1,3)" = model_C
  )
  for (name in names(models)) {

    cat("\n==============================================\n")
    cat("Normality Tests for", name, "\n")

    res <- residuals(models[[name]])

    # Histogram of residuals
```

```
    hist(res,
         main = paste("Histogram of Residuals -", name),
         col = "lightblue",
         breaks = 10)

    # QQ plot
    qqnorm(res, main = paste("QQ Plot -", name))
    qqline(res, col = "red")

    # Shapiro-Wilk normality test
    shapiro_result <- shapiro.test(res)
    print(shapiro_result)
  }
==============================================
Normality Tests for ARIMA(2,1,2)

        Shapiro-Wilk normality test
data:  res
W = 0.99533, p-value = 0.9827
==============================================
Normality Tests for ARIMA(2,1,1)
        Shapiro-Wilk normality test
data:  res
W = 0.99469, p-value = 0.9663
==============================================
Normality Tests for ARIMA(2,1,3)
        Shapiro-Wilk normality test
data:  res
W = 0.99626, p-value = 0.9954
```

To check whether the residuals are normally distributed, I used three methods: histograms, QQ plots and the Shapiro–Wilk test.

1. QQ Plots

From the QQ plots of all three models:

- Most of the points lie very close to the straight reference line.
- There is only a very small deviation in the extreme tails.
- No strong curvature or systematic departure from normality is observed.

Overall, the QQ plots suggest that the residuals are approximately normally distributed.

2. Shapiro–Wilk Test

The results are:

- ARIMA(2,1,2): p-value = 0.9827
- ARIMA(2,1,1): p-value = 0.9663
- ARIMA(2,1,3): p-value = 0.9954

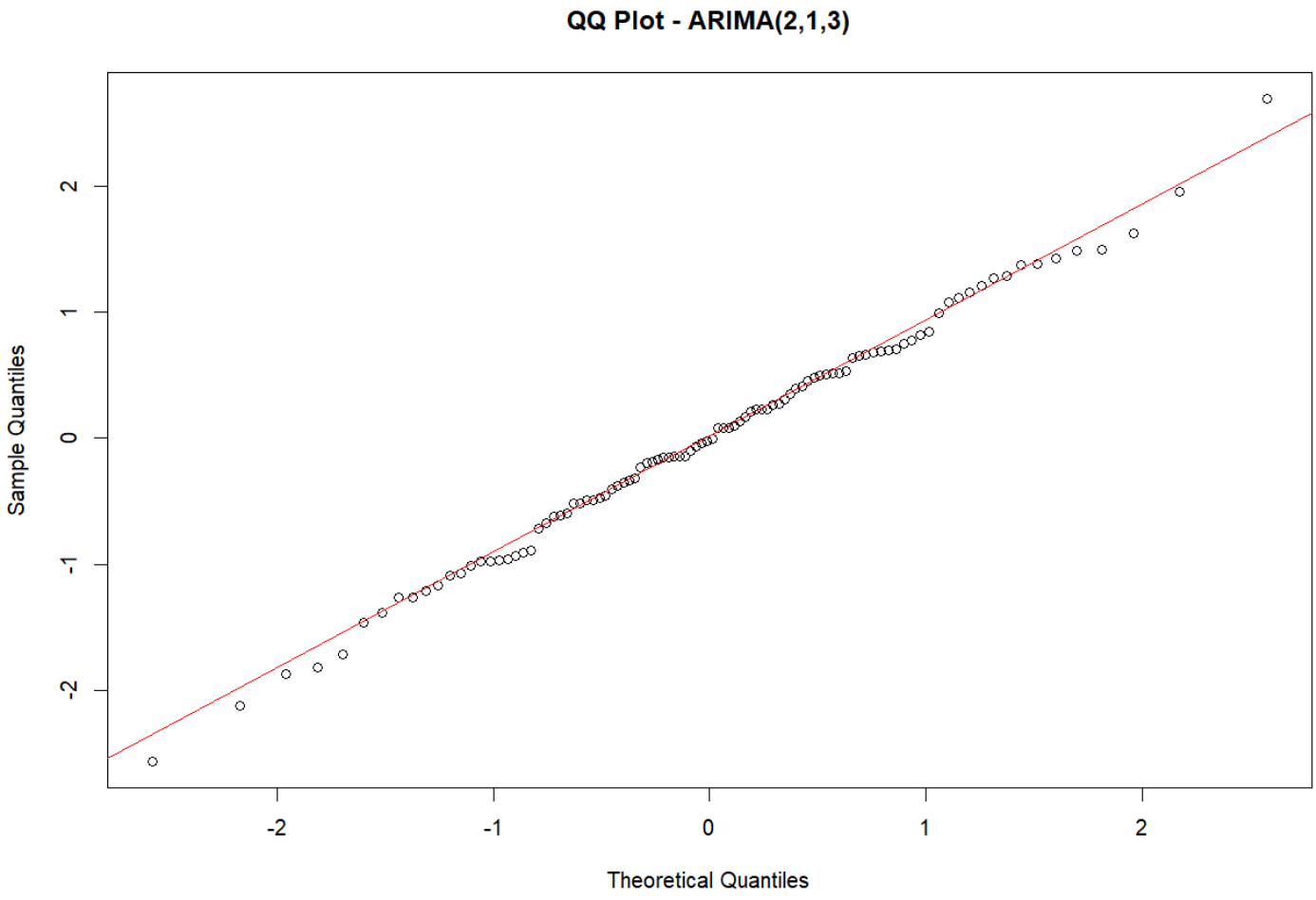
All p-values are much greater than 0.05. Therefore, we fail to reject the null hypothesis of normality for all three models.

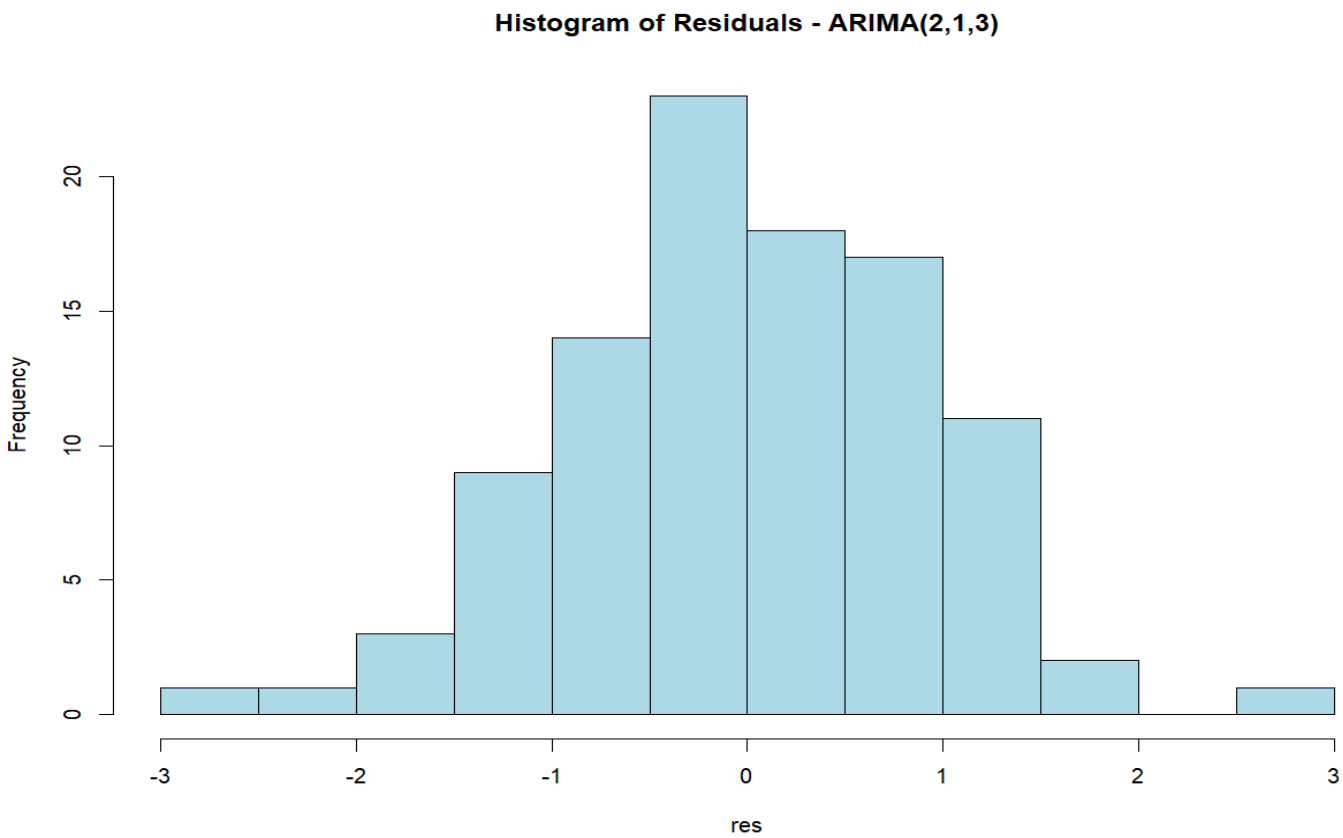
The W statistics are also very close to 1, which indicates a very good fit to normal distribution.

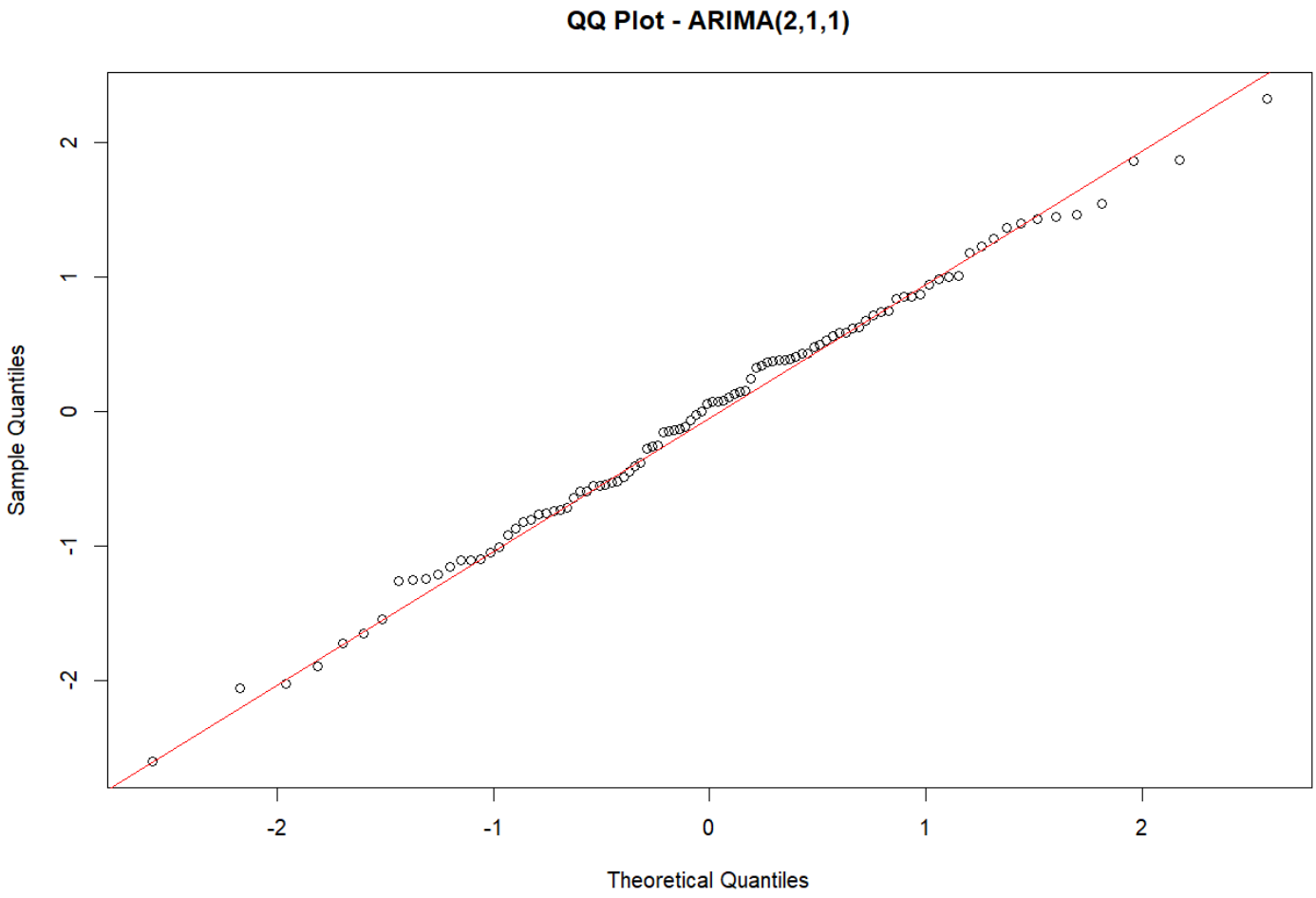
Conclusion:

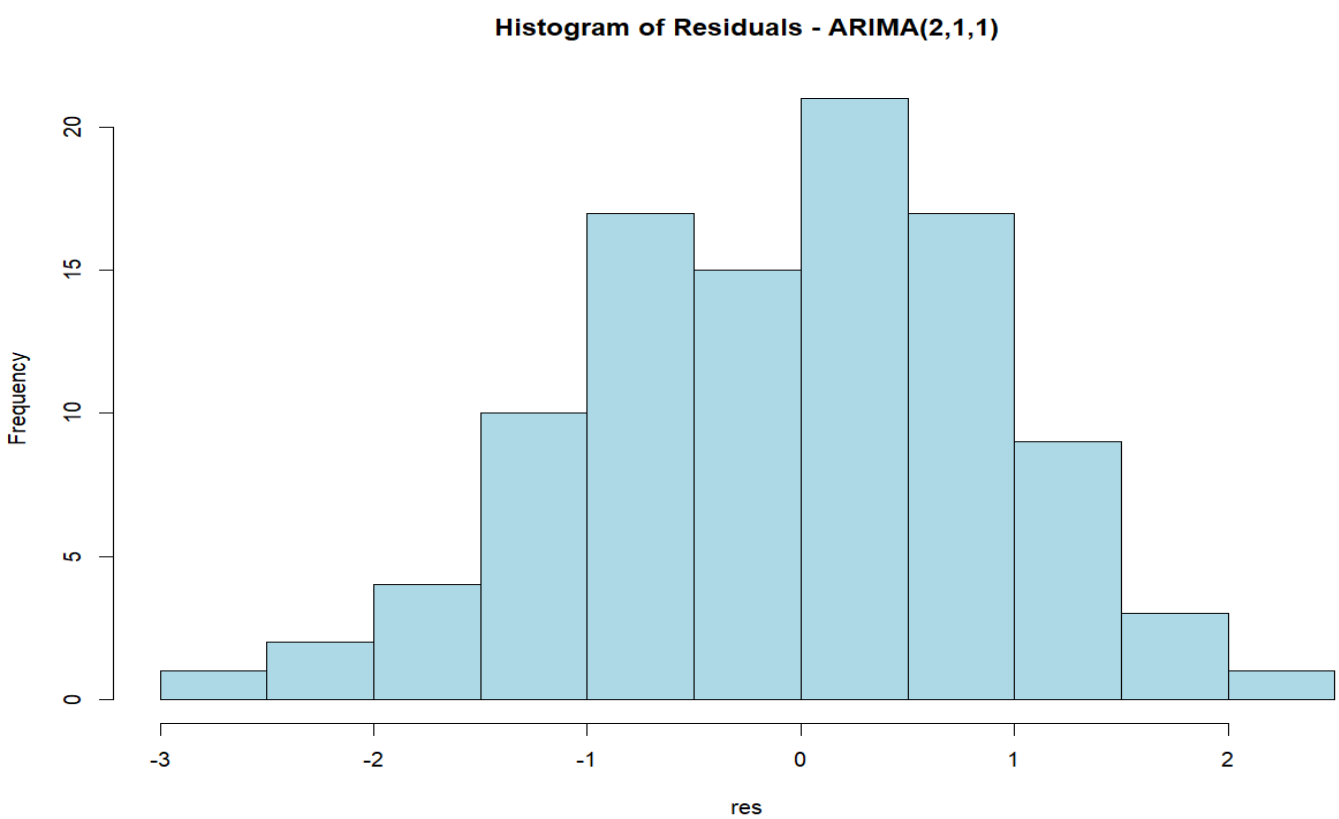
Based on both graphical analysis (QQ plots and histograms) and the Shapiro–Wilk test, there is no statistical evidence against normality of the residuals.

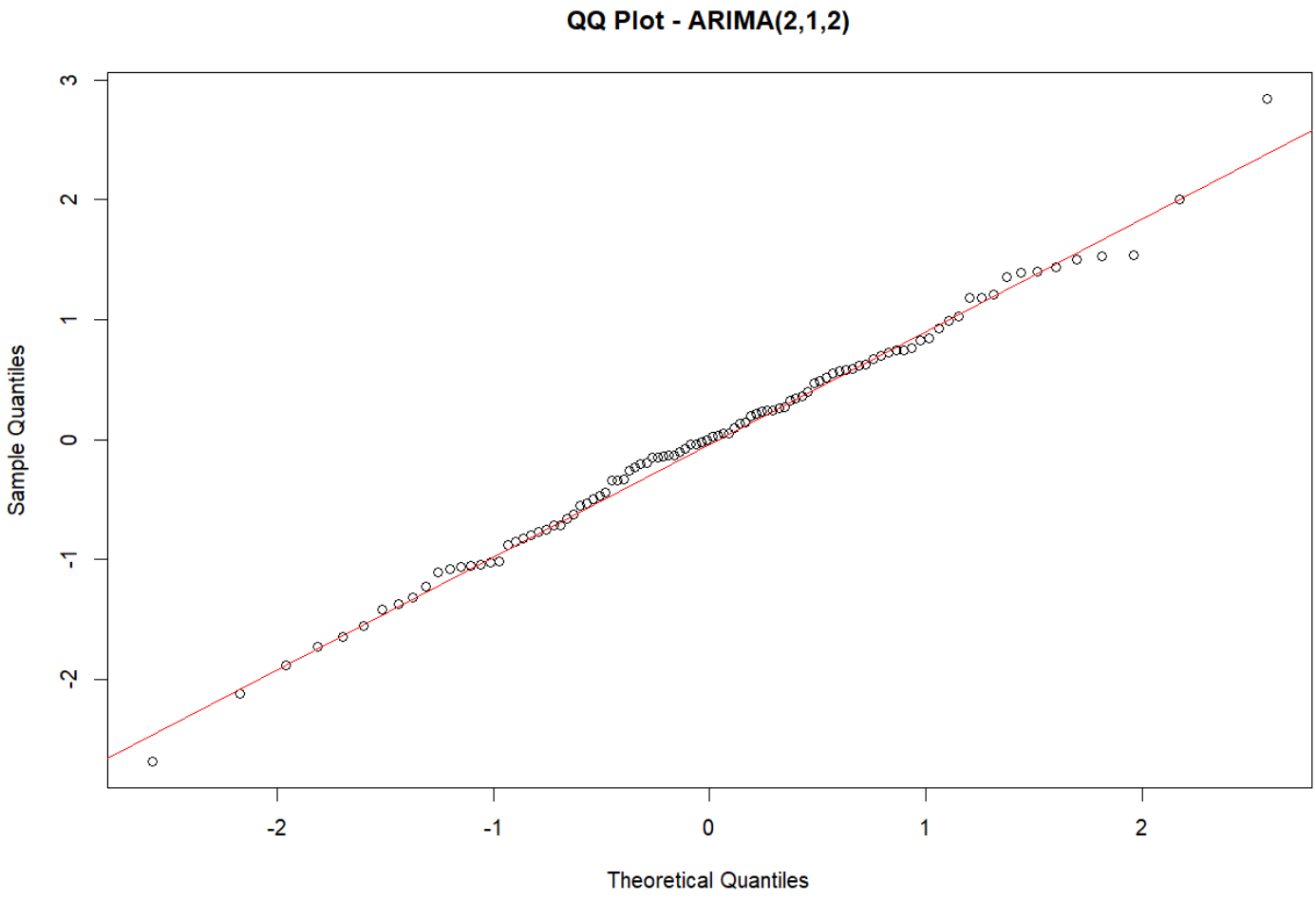
Thus, for all three ARIMA specifications, the residuals can be considered approximately normally distributed.

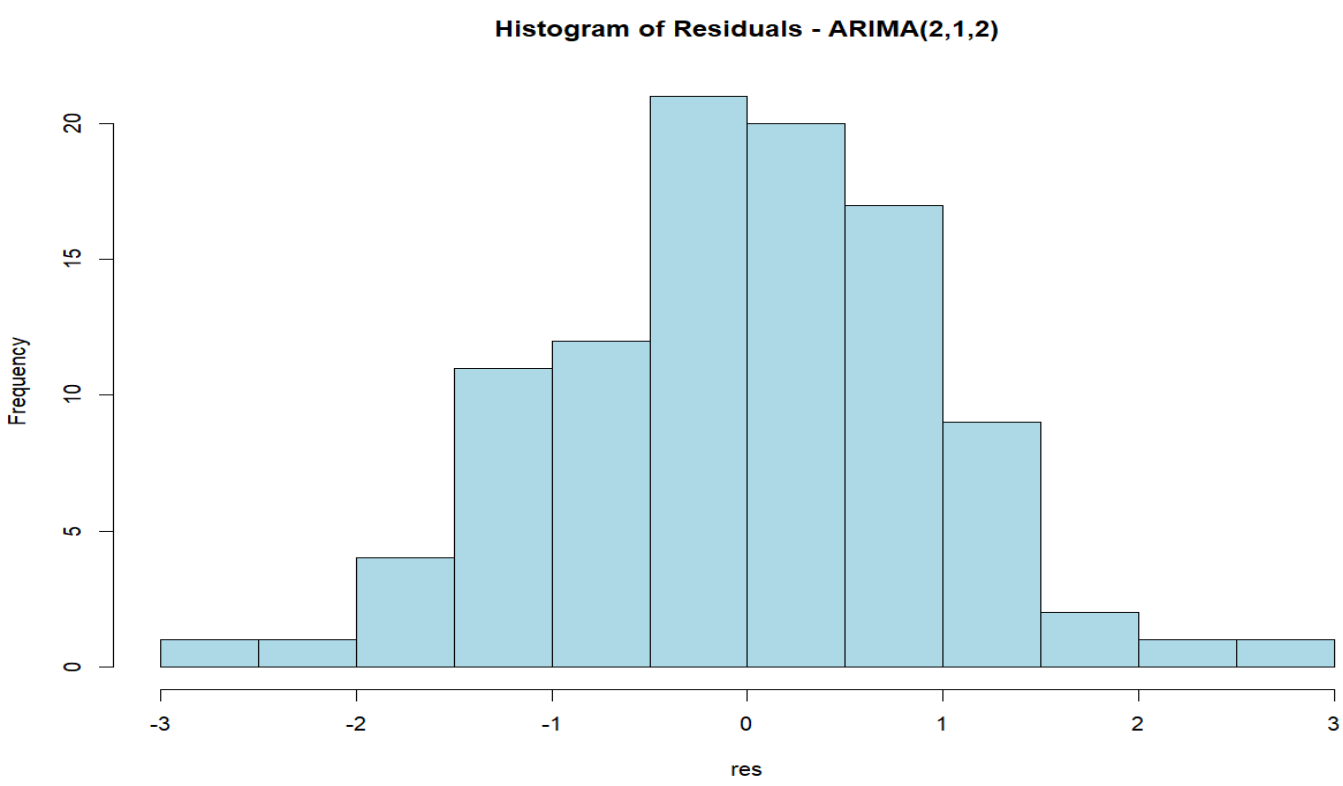

QQ Plot - ARIMA(2,1,3)
Sample Quantiles
Theoretical Quantiles
-2
-1
0
1
2


Histogram of Residuals - ARIMA(2,1,3)
Frequency
res
0
5
10
15
20
-3
-2
-1
0
1
2
3


QQ Plot - ARIMA(2,1,1)
Sample Quantiles
Theoretical Quantiles
-2
-1
0
1
2


Histogram of Residuals - ARIMA(2,1,1)
Frequency
res
0
5
10
15
20
-3
-2
-1
0
1
2


QQ Plot - ARIMA(2,1,2)
Sample Quantiles
Theoretical Quantiles
-2
-1
0
1
2
3


Histogram of Residuals - ARIMA(2,1,2)
Frequency
res
0
5
10
15
20
-3
-2
-1
0
1
2
3

3. Final Model Selection

In order to choose the preferred specification, I consider all previous results including information criteria, diagnostic tests, and the principle of parsimony.

First, according to the information criteria:

- ✓ ARIMA(2,1,2) minimizes the AIC.
- ✓ ARIMA(2,1,1) minimizes the BIC.
- ✓ ARIMA(2,1,3) has slightly higher values of both criteria.

Second, the diagnostic tests show that:

- ✓ For all three models, the Ljung–Box test does not reject the null hypothesis of no autocorrelation.
- ✓ The ACF and PACF of residuals indicate no remaining serial correlation.
- ✓ The Shapiro–Wilk test suggests that residuals are approximately normally distributed in all cases.

Therefore, from a statistical point of view, all three models seem adequate.

However, according to the principle of parsimony, when two models perform similarly, the one with fewer parameters should be preferred, especially with respect to the moving average part.

ARIMA(2,1,1) has:

- ✓ One fewer MA parameter compared to ARIMA(2,1,2)
- ✓ Two fewer MA parameters compared to ARIMA(2,1,3)
- ✓ The lowest BIC (which penalizes complexity more strongly)
- ✓ Although ARIMA(2,1,2) slightly improves the AIC, the improvement is relatively small.

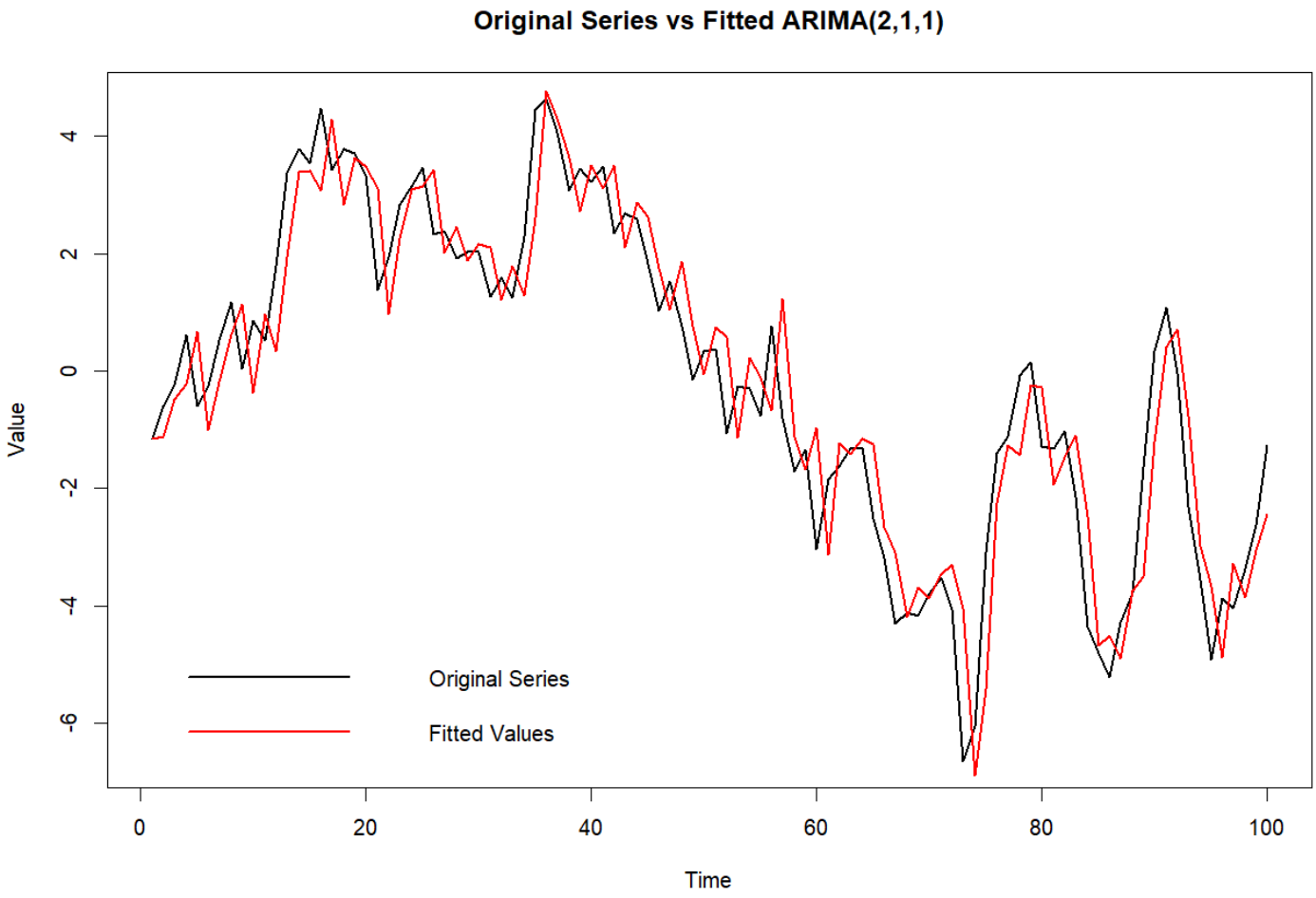
Taking everything into account, I choose:

ARIMA(2,1,1)

as the preferred specification, since it provides a good fit, passes all diagnostic checks, and is more parsimonious compared to the competing models.

4. Plot original time series and best-fitted model.

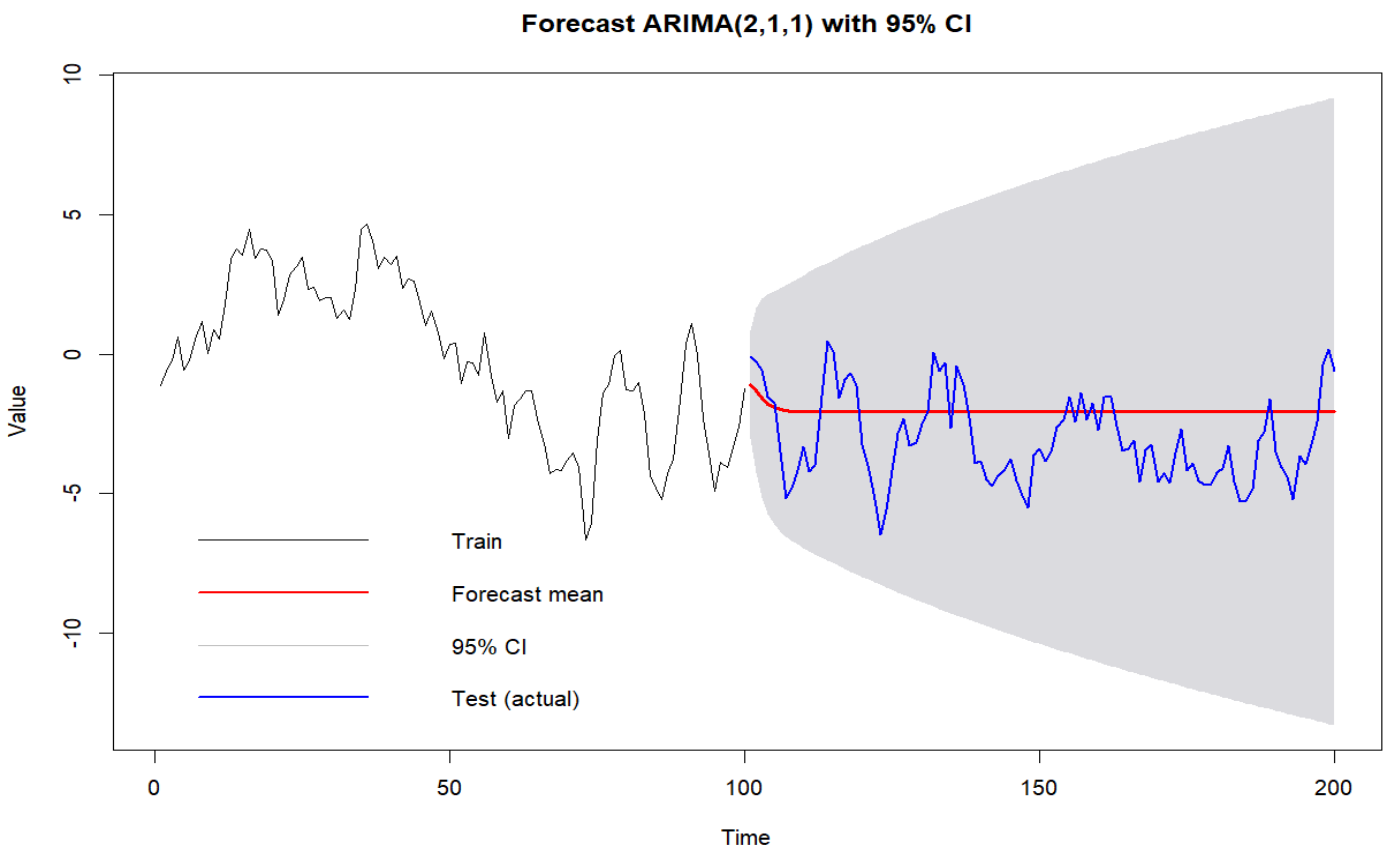
```
# Final selected model
  best_model <- model_B   # ARIMA(2,1,1)
# Extract fitted values
  fitted_values <- fitted(best_model)
# Plot original series
  plot(yt_series,
       type = "l",
       col = "black",
       lwd = 2,
       main = "Original Series vs Fitted ARIMA(2,1,1)",
       ylab = "Value",
       xlab = "Time")
# Add fitted values
  lines(fitted_values,
        col = "red",
        lwd = 2)
 legend("bottomleft",
         legend = c("Original Series", "Fitted Values"),
         col = c("black", "red"),
         lwd = 2,
         bty = "n")
```


Original Series vs Fitted ARIMA(2,1,1)
Value
Time
Original Series
Fitted Values
4
2
0
-2
-4
-6
0
20
40
60
80
100

Step 4: Forecast

1.Generate a forecast plot for a time series, incorporating a 95% confidence interval. Plot the original time series and h–step prediction. Take h = 10, h = 25 and h = 100.

```
# Train/Test split
  Full_Dataset <- read.csv("Case_study.csv")
  train <- ts(Full_Dataset[1:100, 45])
  test  <- ts(Full_Dataset[-(1:100), 45])

# Fit final ARIMA model
  best_fit <- Arima(train, order = c(2,1,1))

# Forecast horizons
  h_values <- c(10, 25, 100)

# Forecast plots
  for (h in h_values) {

    h_use <- min(h, length(test))

    fc <- forecast(best_fit, h = h_use, level = 95)

    # Plot forecast with confidence interval
    plot(fc, main="Forecast ARIMA(2,1,1) with 95% CI",
         xlab="Time", ylab="Value")

    # Highlight forecast mean
    lines(fc$mean, col="red", lwd=3)

    # Plot actual test data
    lines(ts(test[1:h_use], start=101), col="blue", lwd=2)

    legend("bottomleft",
           legend = c("Train", "Forecast mean", "95% CI", "Test (actual)"),
           col = c("black", "red", "gray", "blue"),
           lwd = c(1, 2, 1, 2),
           bty = "n")
  }
```



In this step, the forecasting ability of the selected ARIMA(2,1,1) model was evaluated using out-of-sample data. The first 100 observations were used as the training data and the remaining observations were used as the test set. Forecasts were generated for three horizons (h = 10, 25 and 100) with a 95% confidence interval.

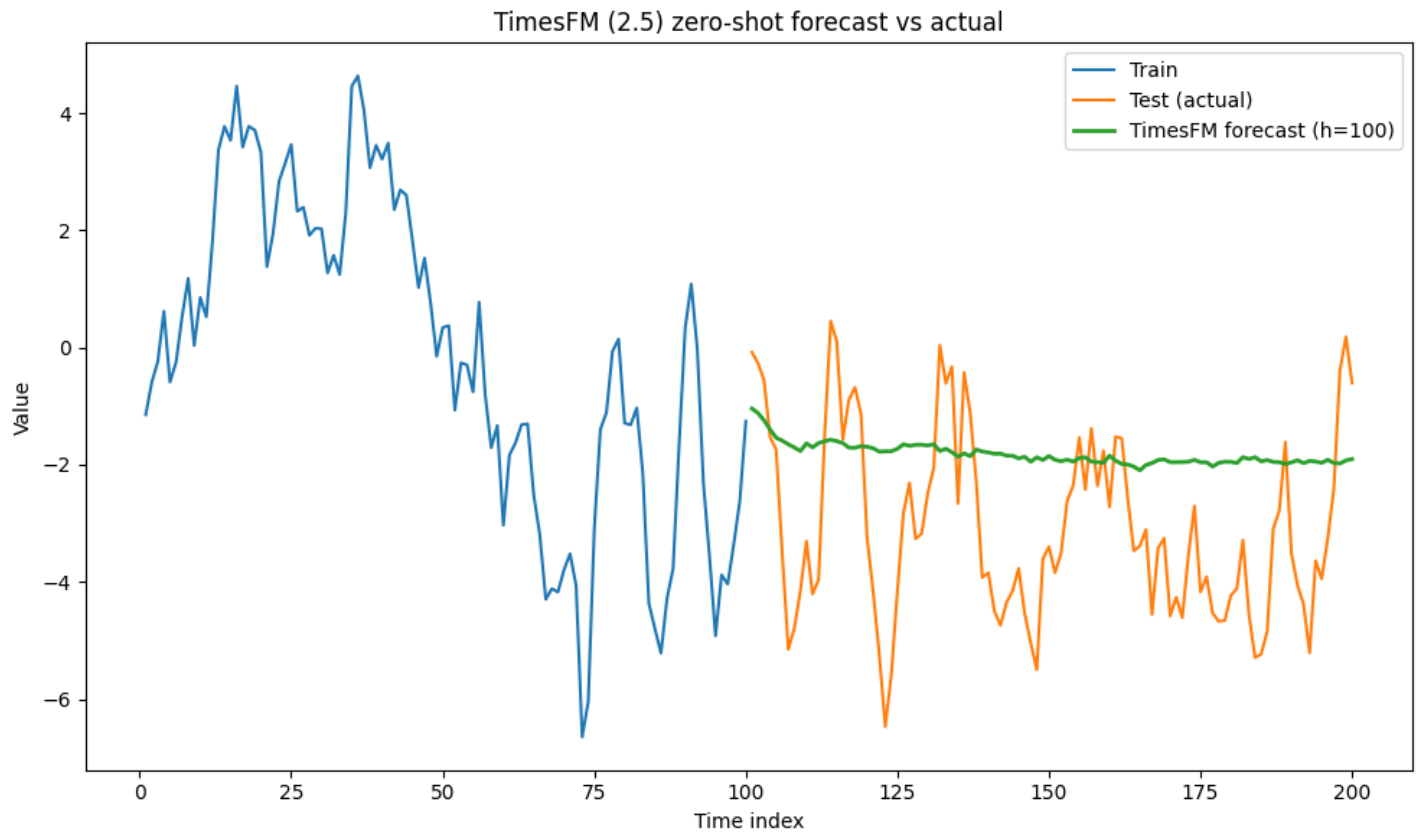
From the forecast plots we can see that the prediction intervals become wider when the forecast horizon increases, which means the uncertainty grows for long-term prediction. The forecast mean also becomes almost flat after some steps, which is typical for ARIMA models

2. Calculate MSE

```
# Compute Mean Squared Error
  mse_results <- data.frame(h = integer(), MSE = double())
  for (h in h_values) {

    h_use <- min(h, length(test))
    fc <- forecast(best_fit, h = h_use, level = 95)

    mse <- mean((as.numeric(test[1:h_use]) - as.numeric(fc$mean))^2)

    mse_results <- rbind(mse_results, data.frame(h = h_use, MSE = mse))
  }
  print(mse_results)
    h      MSE
1  10 2.865076
2  25 4.191318
3 100 3.419277
```

To measure the forecast accuracy, the Mean Squared Error (MSE) was calculated by comparing the predicted values with the actual test observations. The MSE values are 2.865 for h = 10, 4.191 for h = 25 and 3.419 for h = 100.

Optional Bonus: TimesFM Forecast (Compare with ARIMA)



The TimesFM model was used in a zero-shot setting to generate forecasts for the same horizons (h = 10, 25, and 100). The prediction results are shown in the figure where the training data, the actual test data, and the TimesFM forecasts are plotted together.

From the plot it can be seen that the TimesFM forecast mostly stays around a similar level and does not fully follow the ups and downs of the real series. Some turning points in the actual data are missed by the model, specially when the values change quickly. So the general level of the series is captured, but the detailed movements are not always predicted well.

This is also reflected in the MSE values. The TimesFM model produced MSE values of about 3.61, 4.79, and 3.82 for horizons 10, 25 and 100. Compared to the ARIMA model used before, the errors are slightly larger. Overall, the TimesFM model gives a reasonable forecast pattern, but in this case the ARIMA model seems to match the data a bit better.